

Video game

A **video game**,^[a] **computer game**,^[b] or simply **game** is an electronic game that involves interaction with a user interface or input device (such as a joystick, controller, keyboard, or motion sensing device) to generate visual feedback from a display device, most commonly shown in a video format on a television set, computer monitor, flat-panel display or touchscreen on handheld devices, or a virtual reality headset. Most modern video games are audiovisual, with audio complement delivered through speakers or headphones, and sometimes also with other types of sensory feedback (e.g., haptic technology that provides tactile sensations). Some video games also allow microphone and webcam inputs for in-game chatting and livestreaming.



A Pong-clone console of the first generation at the Computerspielemuseum Berlin

Video games are typically categorized according to their hardware platform, which traditionally includes arcade video games, console games, and computer games (which includes LAN games, online games, and browser games). More recently, the video game industry has expanded onto mobile gaming through mobile devices (such as smartphones and tablet computers), virtual and augmented reality systems, and remote cloud gaming. Video games are also classified into a wide range of genres based on their style of gameplay and target audience.

The first video game prototypes in the 1950s and 1960s were simple extensions of electronic games using video-like output from large, room-sized mainframe computers. The first consumer video game was the arcade video game Computer Space in 1971, which took inspiration from the earlier 1962 computer game Spacewar!. In 1972 came the now-iconic video game Pong and the first home console, the Magnavox Odyssey. The industry grew quickly during the "golden age" of arcade video games from the late 1970s to early 1980s but suffered from the crash of the North American video game market in 1983 due to loss of publishing control and saturation of the market. Following the crash, the industry matured, was dominated by Japanese companies such as Nintendo, Sega, and Sony, and established practices and methods around the development and distribution of video games to prevent a similar crash in the future, many of which continue to be followed. In the 2000s, the core industry centered on "AAA" games, leaving little room for riskier experimental games. Coupled with the availability of the Internet and digital distribution, this gave room for independent video game development (or "indie games") to gain prominence into the 2010s. Since then, the commercial importance of the video game industry has been increasing. The emerging Asian markets and proliferation of smartphone games in particular are altering player demographics towards casual and cozy gaming, and increasing monetization by incorporating games as a service.

Today, video game development requires numerous skills, vision, teamwork, and liaisons between different parties, including developers, publishers, distributors, retailers, hardware manufacturers, and other marketers, to successfully bring a game to its consumers. As of 2020, the global video game market had estimated annual revenues of US\$159 billion across hardware, software, and services, which is three times the size of the global music industry and four times that of the film industry in 2019,^[1] making it a formidable heavyweight across the modern entertainment industry. The video game market is also a major influence behind the electronics industry, where personal computer component, console, and peripheral sales, as well as consumer demands for better game performance, have been powerful driving factors for hardware design and innovation.

Origins

Early video games used interactive electronic devices with various display formats. The earliest example dates to 1947—a "cathode-ray tube amusement device" was filed for a patent on 25 January 1947, by Thomas T. Goldsmith Jr. and Estle Ray Mann, and issued on 14 December 1948, as U.S. Patent 2455992.^[2] Inspired by radar display technology, it consisted of an analog device allowing a user to control the parabolic arc of a dot on the screen to simulate a missile being fired at targets, which were paper drawings fixed to the screen.^[3] Other early examples include the Nimrod computer at the 1951 Festival of Britain; Christopher Strachey's *Checkers*, possibly the first game to display visuals on an electronic screen in 1952; *OXO*, a tic-tac-toe computer game by Alexander S. Douglas for the EDSAC in 1952; *Tennis for Two*, an electronic interactive game engineered by William Higinbotham in 1958; and *Spacewar!*, written by Massachusetts Institute of Technology students Martin Graetz, Steve Russell, and Wayne Wiitanen's on a DEC PDP-1 computer in 1962. Each game had different means of display: NIMROD had a panel of lights to play the game of Nim,^[4] *OXO* had a graphical display to play tic-tac-toe,^[5] *Tennis for Two* had an oscilloscope to display a side view of a tennis court,^[3] and *Spacewar!* had the DEC PDP-1's vector display to have two spaceships battle each other.^[6]

These inventions laid the foundation for modern video games. In 1966, while working at Sanders Associates, Ralph H. Baer devised a system to play a basic table tennis game on a television screen. With the company's approval, Baer created the prototype known as the "Brown Box". Sanders patented Baer's innovations and licensed them to Magnavox, which commercialized the technology as the first home video game console, the Magnavox Odyssey, released in 1972.^{[7][8]} Separately, Nolan Bushnell and Ted Dabney,



Baer in 2009



Bushnell in 2013

Ralph H. Baer and Nolan Bushnell have each been described as the "Father of Video Games" for their work in the field



Tennis for Two (1958), an early analog computer game that used an oscilloscope for a display